

Project — Estimate the Mean Voltage from Noisy Recordings

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given voltage recordings from 6 sessions. Each session has irregular sample times (samples come at random intervals). Your job is to estimate the mean voltage V_{ss} for each session and check how the estimate depends on how densely the session was sampled.

Background (short)

The neuron's voltage fluctuates around a steady-state value V_{ss} :

$$V(t) = V_{ss} + \text{noise}$$

Estimating V_{ss} is simple: just take the mean of all samples. But the standard error of the estimate depends on how many samples you have.

Data

In `shared/data/`:

- `voltage_sessions.npz` — `session_id`, `time_ms`, `V_obs_mV` for every sample.
- `session_protocol.csv` — for each session: mean sample interval, number of samples, role (fit or holdout).

Steps

1. Load and look. Open the starter notebook. Load the sessions and the protocol. Print the number of samples per session.
2. Plot one session. Plot `V_obs` vs. `time_ms` for session 0. Notice the irregular sample spacing.
3. Estimate V_{ss} for each session. For each session, compute the mean and standard error of `V_obs`. Store in a table.
4. Standard error vs sample count. Plot standard error against $1 / \sqrt{n_samples}$. A line means the estimator is well-behaved.

What to hand in

- Your filled notebook.

- The plot of one session (voltage vs time).
- The table of V_{ss} estimates and their standard errors.
- The standard-error vs $1/\sqrt{n}$ plot.
- A short paragraph (3-4 sentences) on how the estimate depends on sample count.

If you finish early: compute the same V_{ss} on the first half vs. second half of each session; do the two halves agree?